328654 (28)

BE (6th Semester)
Examination, April - May, 2016

(New Scheme)

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VLSI Design

Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

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Note : Part (*a*) is compulsory in each question, carrying 2 marks. Attempt any **two** parts from (*b*), (*c*) and (*d*), carrying 7 marks each.

1. (*a*) Define the terms SSI, LSI, VLSI and ULSI.

(*b*) What do you mean by VLSI design flow?

(*c*) What do you mean by integrated circuit, a term 'scale of integration' signifies what in it? Write the advantages of the ICs over discrete component.

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(*d*) What are the classifications of ICs? Define the characteristic terms such as fall time, rise time, propagation delay power dissipation in context with the CMOS IC.

2. (*a*) Why CMOS is preferred instead of nMOS and pMOS?

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(*b*) What are the CMOS design rules? Discuss the n-Well fabrication process of CMOS.

(*c*) Write short notes on the stick diagram and little elaborate its feature.

(*d*) What do you mean by twine tub fabrication process of CMOS?

3. (*a*) Draw the circuit diagram of a NOR based ROM array.

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(*b*) What do you mean by CMOS layout design?

(*c*) With the help of a circuit diagram explain the operation of 6-transistor SRAM cell.

(*d*) Design a generic RAM having 16 locations and each location can store 8 bits of data.

4. (*a*) What are the different delays of VHDL?

(*b*) Write a VHDL code to design a sequence detector that detects of sequence "101" from the data arriving at the port 'w' serially. The circuit is positive edge triggered.

(*c*) Explain the internal structure of CPLD. How the output is considered to be the registered output?

(*d*) Write the difference between the FPGA and CPLD.

5. (*a*) Write a VHDL code to generate the stimulus "11101001". The sequence must be non-repetitive.

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(*b*) Write VHDL code for the 4-bit up down counter.

(*c*) Write short note on Moore's finite state machine, also explain the FSM in terms of VHDL.

(*d*) Write the difference between Mealy and Moore's FSM model along with their advantages.

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